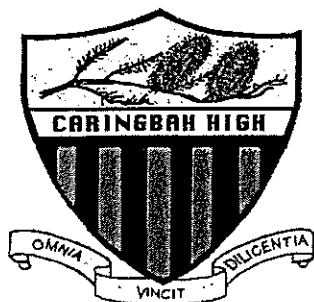




2017 Year 11
Semester 2 Exam

Mathematics (2 Unit)

General Instructions

- Approved calculators may be used.
- Start each question in a new booklet.
- Answers not supported with working may not attract full marks.
- Marks may not be awarded for carelessly arranged work.
- Write using a black or blue pen
- A reference sheet with mathematical formulae is provided

Time Allowed

2 hours + 3 minutes reading time

Answer questions 1-5 on the multiple choice answer sheet provided

Multiple Choice (1 Mark each)

Question 1

If $x = a\left(b - \frac{1}{y}\right)$ then

A) $y = \frac{a}{b-x}$

B) $y = \frac{a}{ab-x}$

C) $y = \frac{1}{ab-x}$

D) $y = \frac{x}{a} - b$

Question 2

The solution to the inequality

$$6 - x - x^2 \leq 0 \text{ is}$$

A) $-3 \leq x \leq 2$

B) $x \leq -2$ or $x \geq 3$

C) $x \leq -3$ or $x \geq 2$

D) $-2 \leq x \leq 3$

Question 3

What is the perpendicular distance between the lines $y = x + 5$ and $x - y = 3$

A) $4\sqrt{2}$

B) 4

C) $\sqrt{2}$

D) $2\sqrt{2}$

Question 4

Which of the following trigonometric expressions is equivalent to $\tan(90 - x)^\circ$?

- A) $\tan x^\circ$
- B) $\cot x^\circ$
- C) $-\tan x^\circ$
- D) $-\cot x^\circ$

Question 5

The midpoint of $A(x, 2), B(-5, y)$ is $M(8, -10)$

The value of x, y is,

- A) $x = -21, y = 22$
- B) $x = 21, y = 22$
- C) $x = -21, y = -22$
- D) $x = 21, y = -22$

Question 6 (12 Marks)

Marks

- a) Find the value of y such that

$$2\sqrt{50} - \sqrt{72} = \sqrt{y} \quad (2)$$

- b) Express with a rational denominator

$$\frac{\sqrt{2}}{\sqrt{3}+1} \quad (2)$$

- c) Solve the equation $2\cos\theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$ (2)

- d) Find $\lim_{x \rightarrow -4} \frac{x^2 + x - 12}{x + 4}$ (2)

- e) Graph the region ($\frac{1}{3}$ page size) where the inequations hold simultaneously
 $y > x^2$ and $y \geq x + 6$ (3)

- f) Show that $\cos\theta \tan\theta = \sin\theta$ (1)

Question 7 (12 Marks)**Marks**

- a) State the domain and range of $y = \sqrt{x-2}$ (2)

b) If $f(x) = \begin{bmatrix} x^2 - 1 & x \geq 1 \\ 2 - x & -1 < x < 1 \\ \frac{1}{x} & x \leq -1 \end{bmatrix}$

Find the value of $2f(3) - f(-3)$ (2)

- c) Simplify $\frac{16}{2^{3x} \times 8^{1-x}}$ (2)

- d) Draw separate sketches ($\frac{1}{3}$ page size) of the following. Show any intercepts with the axes, and any asymptotes.

i) $y = 2x - 1$ (1)

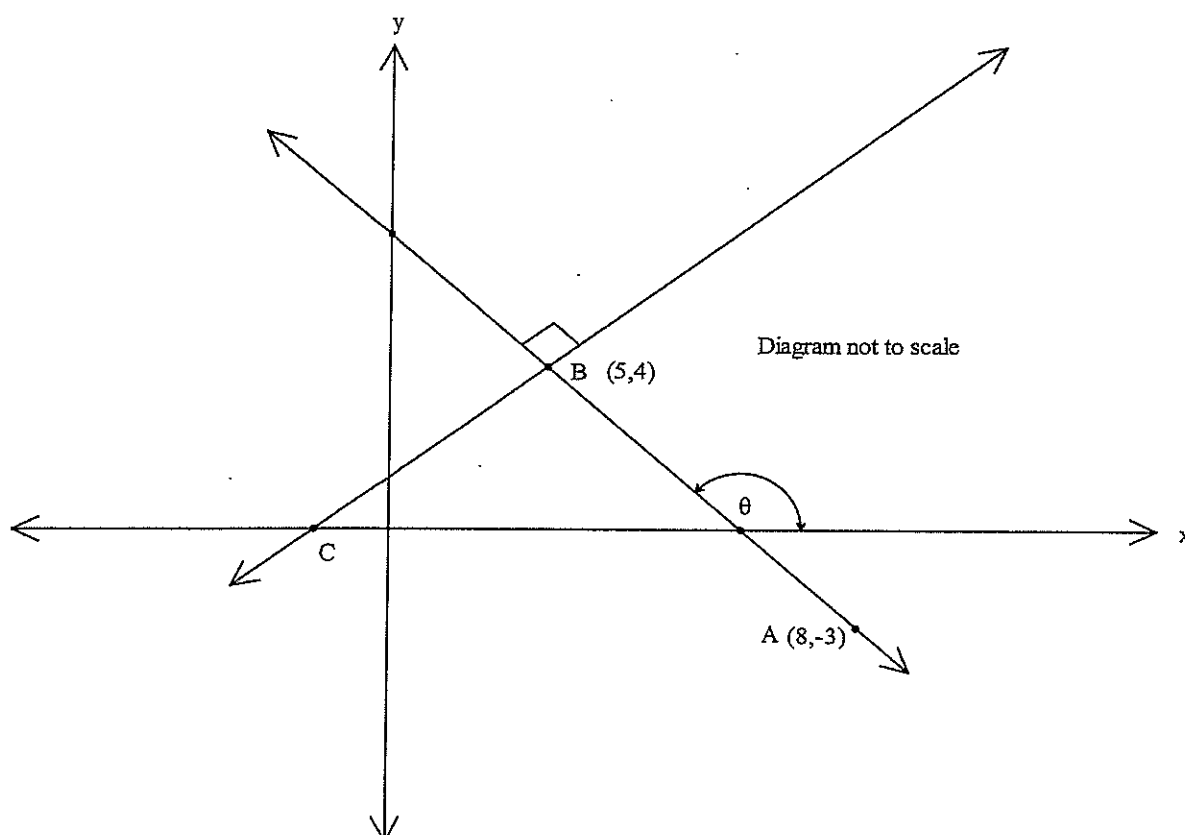
ii) $y = -\sqrt{4 - x^2}$ (2)

iii) $y = 2^x + 1$ (3)

Question 8 (12 Marks)

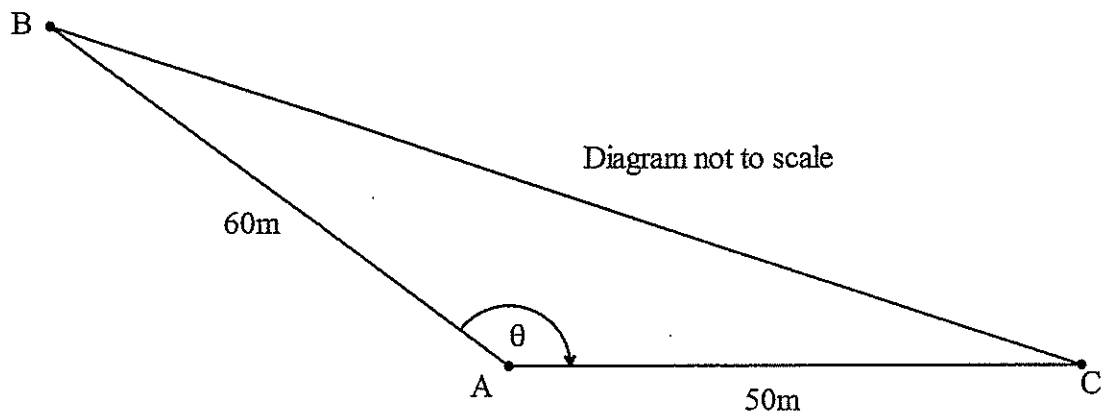
Marks

- a) The points $A(8, -3)$ and $B(5, 4)$ are shown in the diagram below. The line through AB makes an angle of θ with the positive x axis and the point C lies on the x axis.



- i) Find the gradient of the line AB (1)
- ii) Find the value of θ to the nearest degree. (1)
- iii) Find the coordinates of C given $AB \perp BC$ (2)
- iv) Find the coordinates of M , the midpoint of AB (1)
- v) Find the equation of the line through M parallel to the line BC in general form (2)

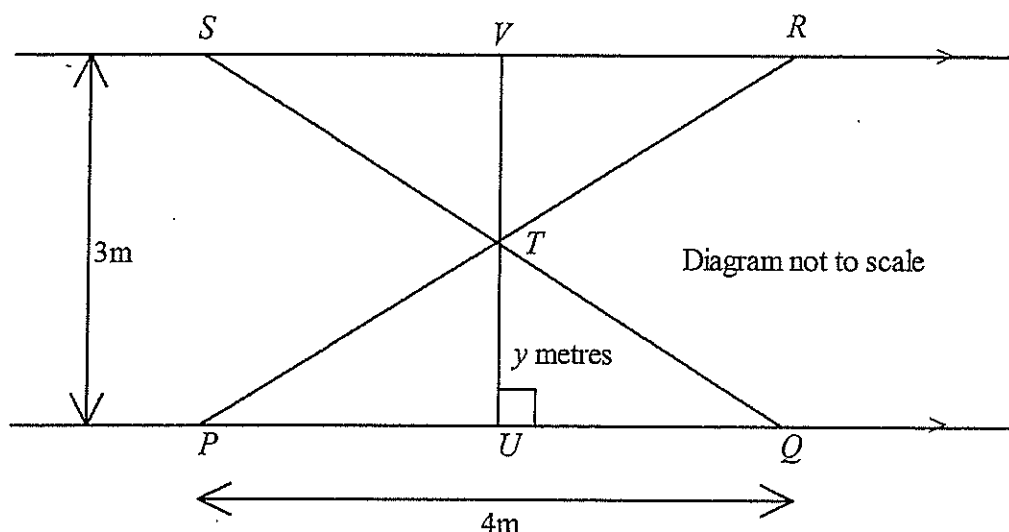
- b) Two sides of a triangular field are $AB = 60$ m, $AC = 50$ m and the included angle $\angle BAC = \theta$ is obtuse.



- i) If the area of the triangle is 750m^2 , find the size of $\angle BAC$ (2)
- ii) Find the length of the side BC (1 decimal place) (2)
- c) Make b the subject if $\frac{1}{x} = \frac{1}{a} + \frac{1}{b}$ (1)

Question 9 (11 Marks)

Marks



a) In the diagram PQ and SR are parallel railings 3metres apart. The points P and Q are fixed 4 metres apart on the lower railing. Two crossbars PR and QS intersect at T as shown in the diagram. The line through T perpendicular to PQ intersects PQ at U and SR at V . The length UT is y metres.

i. By using similar triangles, or otherwise, show that $\frac{SR}{PQ} = \frac{VT}{UT}$ (3)

ii. Show that $SR = \frac{12}{y} - 4$ (2)

iii. Find the total area, A of $\triangle PTQ$ and $\triangle RTS$ in terms of y in simplest form (3)

b) Solve for x

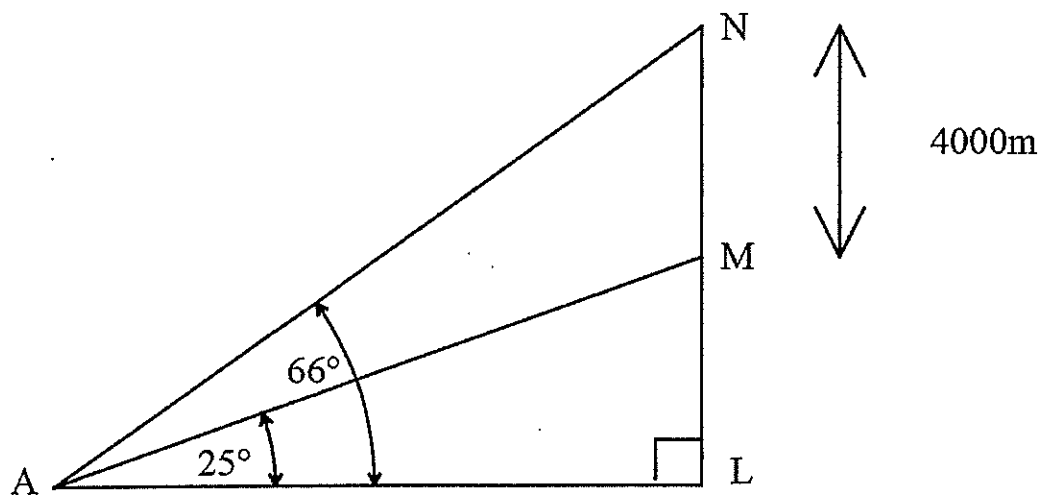
$$\frac{1}{2\sqrt{5} + \sqrt{x}} + \frac{1}{2\sqrt{5} - \sqrt{x}} = 2\sqrt{5}$$

(3)

Question 10 (11 Marks)

Marks

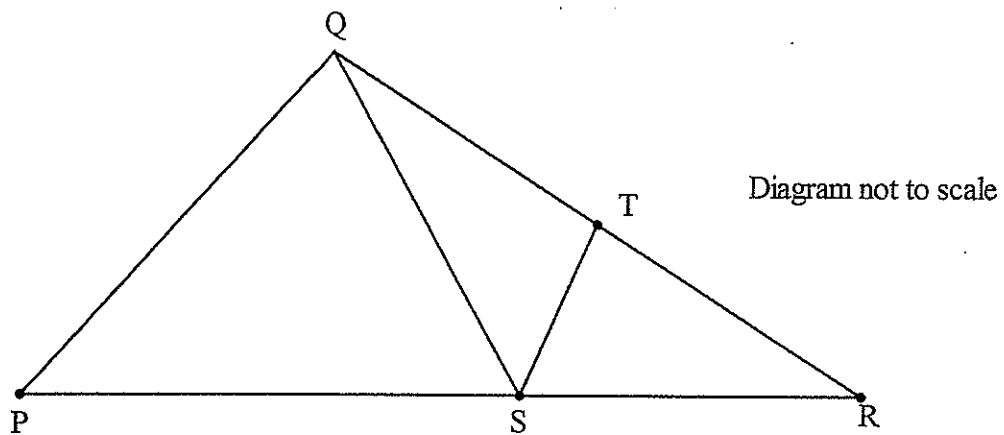
- a) A rocket launched vertically from L is observed on from A . Soon after launch when at position M its angle of elevation is 25° . After it climbs 4000 metres from this position to N its angle of elevation is 66°



- i) Find $\angle ANL$ (1)
- ii) Find length of AM (to nearest metre) (2)
- iii) Find how far the observer is from the launch pad (to nearest metre) (2)

Question 10 (continued)

b)



In $\triangle PQR$, T lies on side QR and S lies on side PR such that
 $QT = TR$, $QS = QP$, $ST \perp QT$

- i) Copy the diagram into your answer booklet and show all the given information. (1)
- ii) Prove that $\triangle QTS \cong \triangle RTS$ (2)
- iii) Prove that $\angle QPS = 2\angle TQS$ (3)

Question 11 (11 Marks)**Marks**

a. Consider the circle $(x-1)^2 + (y-2)^2 = 18$

i) Find the centre of the circle and its radius (2)

ii) Show that the line $y = x - 5$ is tangent to the circle (3)

b. Consider the function $g(x) = \frac{2x}{x^2 - 1}$

i) State the domain of this function (1)

ii) Determine with justification whether the function is odd, even, or neither (2)

c. Given $f(x) = \sqrt{5x-25} - \sqrt{x-1}$ find the value of x for which $f(x) = 2$. (3)

Question 12 (11 Marks)

Marks

- a. A regular polygon has an exterior angle of 20° .
How many sides does the polygon have? (1)
- b. Prove that $\cos^2(90^\circ - \theta) \cot \theta = \sin \theta \cos \theta$ (2)
- c. Show that $\lim_{x \rightarrow \infty} \frac{4x^2 - x^3 + 2}{3x^3 - x^2 + 1} = \frac{-1}{3}$ (2)
- d. Find the equation of the line passing through the point of intersection of the lines
 $4x - 2y + 3 = 0$ and $x + 4y + 6 = 0$ and which has a gradient of $\frac{2}{3}$. (3)
- e. A large tank can be filled by 2 similar small pumps and 1 larger pump working together in
1 hour and 12 minutes. The larger pump B alone takes 1 hour less than the smaller pump A
alone to fill the tank. Find out how long each pump takes, given that $\frac{V}{P} = T$ where
 V = Volume of the tank, P = Pump, T = Time (3)

END OF EXAM

Question 6

a) $2\sqrt{50} - \sqrt{72} = \sqrt{y}$

$10\sqrt{2} - 6\sqrt{2} = \sqrt{y}$

$4\sqrt{2} = \sqrt{y}$

$\therefore y = 32$

b) $\frac{\sqrt{2}}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1}$

$= \frac{\sqrt{6} - \sqrt{2}}{2}$

c) $2\cos\theta = 1$

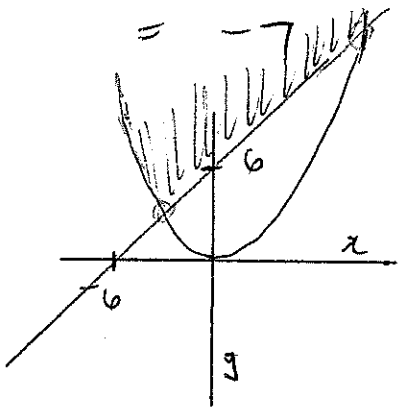
$\cos\theta = \frac{1}{2}$

$\theta = 120^\circ, 240^\circ$

d) $\lim_{x \rightarrow -4} \frac{x^2 + x - 12}{x + 4}$
 $= \frac{(x+4)(x-3)}{x+4}$

$= -7$

e)



f) $\cos\theta \tan\theta = \sin\theta$

L.H.S

$\cancel{\cos\theta} \cdot \frac{\sin\theta}{\cancel{\cos\theta}}$

$= R.H.S.$

Question 7

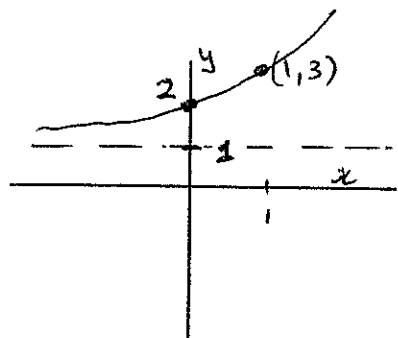
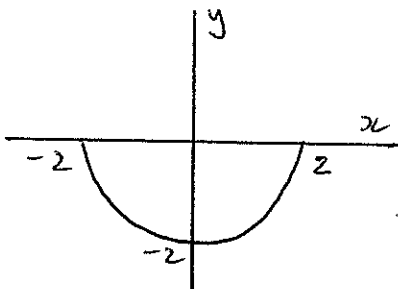
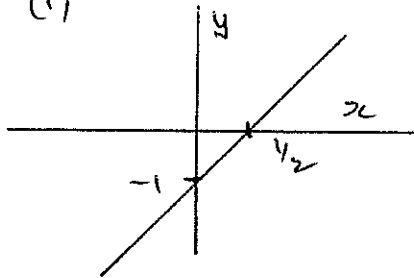
a) $x \geq 2$

b) $2(8) - \frac{1}{3}$
 $= 16\frac{1}{3}$

c) $\frac{16}{2^{32} \times 8^{1-2}}$
 $= \frac{2^4}{2^{32} \times 2^{3-32}}$
 $= \frac{2^4}{2^3}$

$= 2$

d) (i)



Multiple Choice

Q1 B

2 C

3 A

4 B

5 D

Question 8

(a) (i) $-\frac{7}{3}$

(ii) $\tan^{-1}(-\frac{7}{3})$
 $= 113^\circ$

(iii) $B(5, 4) \ C(x, 0)$

$$\frac{4}{5-x} = \frac{3}{7}$$

$$28 = 15 - 3x$$

$$x = -\frac{13}{3}$$

(iv) $M(1\frac{1}{2}, \frac{1}{2})$

(v) $y-4 = -\frac{7}{3}(x-5)$

$$3y-12 = -7x+35$$

$$7x+3y-47=0$$

(b) (i) $750 = \frac{1}{2} \times 60 \times 50 \times \sin \theta$
 $\frac{1}{2} = \sin \theta$

$$\theta = 150^\circ$$

(ii) $BC^2 = 60^2 + 50^2 - 2 \times 60 \times 50 \cos 150^\circ$

$$BC \div 106.3$$

(c) $\frac{1}{x} = \frac{1}{a} + \frac{1}{b}$

$$\frac{1}{x} - \frac{1}{a} = \frac{1}{b}$$

$$\frac{a-x}{ax} = \frac{1}{b}$$

$$b = \frac{ax}{a-x}$$

Question 9

(a) (i) In Δ 's SRT & PQT

$$\angle RST = \angle TQP \text{ (alt } \angle\text{'s, } SR \parallel PQ)$$

$$\angle STR = \angle PTQ \text{ (vert opp } \angle\text{'s)}$$

$$\Delta SRT \parallel \Delta PQT \text{ (equiangular)}$$

hence corresponding heights and sides are in the same ratio.

$$\therefore \frac{SR}{PQ} = \frac{VT}{UT}$$

(ii) $SR = \frac{VT \times PQ}{UT}$

$$= \frac{(3-y) \times 4}{y}$$

$$= \frac{12-4y}{y}$$

$$= \frac{12}{y} - 4$$

(iii) $A = \frac{1}{2} SR \times VT + \frac{1}{2} PQ \times UT$

$$= \frac{1}{2} \left(\frac{12}{y} - 4 \right) (3-y) + \frac{1}{2} (4)(y)$$

$$= \frac{1}{2} \left(\frac{36}{y} - 12 - 12 + 4y \right) + 2y$$

$$= \frac{18}{y} - 12 + 2y + 2y$$

$$= \frac{18}{y} - 12 + 4y$$

Question 9 cont'd

$$(b) \frac{1}{2\sqrt{5} + \sqrt{2}} + \frac{1}{2\sqrt{5} - \sqrt{2}} = 2\sqrt{5}$$

$$\frac{2\sqrt{5} - \sqrt{2}}{20 - 2} + \frac{2\sqrt{5} + \sqrt{2}}{20 - 2} = 2\sqrt{5}$$

$$4\sqrt{5} = 40\sqrt{5} - 2\sqrt{5}x$$

$$-36\sqrt{5} = -2\sqrt{5}x$$

$$x = 18$$

Question 10.

(a) (i) $\angle ANL = 24^\circ$

(ii) $\frac{AM}{\sin 24^\circ} = \frac{4000}{\sin 41^\circ}$

$$AM = \frac{4000 \sin 24^\circ}{\sin 41^\circ}$$

$$AM = 2480 \text{ m.}$$

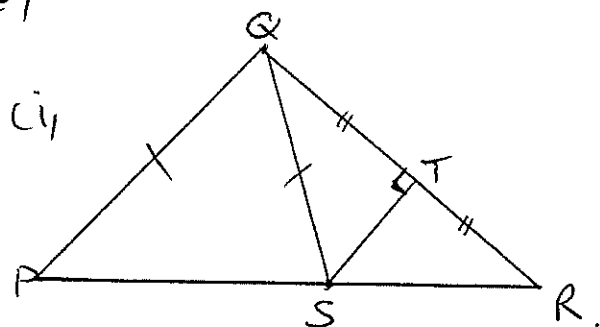
(iii) $\cos 25^\circ = \frac{AL}{AM}$

$$\cos 25^\circ = \frac{AL}{2480}$$

$$AL = 2480 \cos 25^\circ$$

$$AL = 2248 \text{ m.}$$

(b)



(ii) $QT = TR$ given
 $\angle QTS = \angle STR = 90^\circ$ (given)
 ST common side
 $\triangle QTS \equiv \triangle RTS$ (S.A.S.)

(iii) $QS = SR$ (corresponding sides in $\equiv$ triangles from (i))

Let $\angle TSR = \phi$
 $\angle TSQ = \phi$ } corresponding angles in $\equiv$ triangles

$\angle SQT = 90 - \phi$ base angles of $\triangle QSR$

$\angle QSP = 180 - 2\phi$ supplement of $\angle QSR$

$\angle QSP = \angle QPS$ base angles of $\triangle QPS$

$\therefore \angle QPS = 2\angle TQS$

Question 11

(a) $(x-1)^2 + (y-2)^2 = 18$

(i) $C(1,2)$ $r = 3\sqrt{2}$

(ii) $(1,2)$ $x - y - 5 = 0$

$$PD = \frac{|1 - 2 - 5|}{\sqrt{2}}$$

$$= \frac{6}{\sqrt{2}}$$

$$= \frac{6\sqrt{2}}{2}$$

$$= 3\sqrt{2}$$

$\therefore$ equal to radius and $\perp$
thus $y = x - 5$ is tangent
to the circle

(b) (i) $g(x) = \frac{2x}{x^2 - 1}$

all $\mathbb{R}x$, $x \neq \pm 1$

(ii) $g(-x) = \frac{-2x}{x^2 - 1}$

$$g(x) = -g(-x)$$

$\therefore$ odd function

(c)

$$f(x) = \sqrt{5x-25} - \sqrt{x-1}$$

$$2 = \sqrt{5x-25} - \sqrt{x-1}$$

$$4 = 5x - 25 - 2\sqrt{(5x-25)(x-1)} + x - 1$$

$$30 - 6x = -2\sqrt{(5x-25)(x-1)}$$

$$(3x-15)^2 = 5x^2 - 30x + 25$$

$$4x^2 - 10x + 200 = 0$$

$$x^2 - 15x + 50 = 0$$

$$(x-5)(x-10) = 0$$

$$x = 5, x = 10$$

Soln $x = 10$ only.

Question 12

$$(a) \frac{360}{20} \\ = 18$$

$$(b) \cos^2(90-\theta) \cot \theta = \sin \theta \cos \theta$$

$$\text{L.H.S } \sin^2 \theta \cdot \frac{\cos \theta}{\sin \theta} \\ \sin \theta \cos \theta$$

$$(c) \lim_{x \rightarrow \infty} \frac{4x^2 - x^3 + 2}{3x^3 - x^2 + 1}$$

$$\frac{\frac{4x^2}{x^3} - \frac{x^3}{x^3} + \frac{2}{x^3}}{\frac{3x^3}{x^3} - \frac{x^2}{x^3} + \frac{1}{x^3}}$$

$$\frac{\frac{4}{x} - 1 + \frac{2}{x^3}}{3 - \frac{1}{x} + \frac{1}{x^3}}$$

now

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$

$$\frac{0 - 1 + 0}{3 - 0 + 0} \\ = -\frac{1}{3}$$

$$(d) 4x - 2y + 3 + K(x + 4y + 6) = 0$$

$$4x - 2y + 3 + Kx + 4Ky + 6K = 0$$

$$x(4+K) - y(2-4K) + 3+6K = 0$$

$$y = \frac{x(4+K)}{2-4K} + \frac{3+6K}{2-4K}$$

(d) cont'd

$$\frac{4+K}{2-4K} = \frac{2}{3}$$

$$12+3K = 4-8K$$

$$11K = -8$$

$$K = -\frac{8}{11}$$

Eqn of line

$$4x - 2y + 3 - \frac{8}{11}(x + 4y + 6) = 0$$

$$44x - 22y + 33 - 8x - 32y - 48 = 0$$

$$36x - 54y - 15 = 0$$

$$12x - 18y - 5 = 0$$

$$(e) \frac{V}{2A+B} = \frac{6}{5} \quad - (3)$$

$$\frac{V}{B} = T-1 \quad - (1)$$

$$\frac{V}{A} = T \quad - (2)$$

$$\left. \begin{array}{l} \text{from (1)} \frac{V}{T-1} = B \\ \text{from (2)} \frac{V}{T} = A \end{array} \right\} \text{Sub into (3)}$$

$$\frac{V}{\frac{2V}{T} + \frac{V}{T-1}} = \frac{6}{5}$$

$$\frac{V(T-1)}{V[2(T-1)+T]} = \frac{6}{5}$$

$$5T^2 - 23T + 12 = 0$$

$$(5T-3)(T-4) = 0$$

$$T=4 \quad T=\frac{3}{5}$$

$\therefore$ Pump B = 3 hrs

Pump A = 4 hrs